

Algebra Sparks

The mission that names what you've been doing since Mission 01.

THE BIG QUESTION

What does a letter in a math problem actually stand for?

CAPSTONE PROJECT

The Math Trail

A numbered trail through your own house where every stop is a problem you wrote. Mom walks it, and you mark her work.

FLUENCY GAME

Guess My Rule

One person runs the machine, the other cracks the rule in five inputs or fewer.



AXIOM • MISSION COMMANDER, NINE TOURS

“Brock – you’ve been finding the number in the gap all year. This mission just gives it a letter. Last leg.”

Pre-Assessment

Twelve items, six strands. Anything you already own, we skip.

This is not a test. Nothing here counts for a grade. If you have never seen a letter used in place of a number, say so — that is useful information, not a wrong answer.

STRAND A PATTERNS

A1 What comes next: 4, 9, 14, 19, ____ ?

.....

A2 Say the rule of that pattern in words.

.....

STRAND B MACHINES

B1 A machine multiplies by 3. Put in 7.
What comes out?

.....

B2 The same machine gives out 24.
What went in?

.....

STRAND C TWO-STEP RULES

C1 Rule: $\times 2$ then $+ 5$. Put in 6.

.....

C2 Does $+ 5$ then $\times 2$ give the same
answer? Show both.

.....

STRAND D LETTERS

D1 If $n = 5$, what is $3n$?

.....

D2 If $n = 5$, what is $2n + 4$?

.....

STRAND E EQUATIONS

E1 Find x : $x + 6 = 14$

.....

E2 Find x : $5x = 35$

.....

STRAND F MEANING

F1 What does the equals sign claim
about the two sides?

.....

F2 Is $2n$ the same as n^2 ? Test both at $n = 5$.

.....

Skip Chart

Score the pre-assessment, then cross days off the four-week calendar.

2 / 2 → SKIP

1 / 2 → COMPRESS

0 / 2 → TEACH

STRAND	ITEMS	COVERS	SCORE	IF 2/2, REMOVE
A • Patterns	A1 A2	Week 1 • lesson 1.1	--- / 2	The Warm-Up and Core of 1.1. Keep the Challenge — the nth-term work is the whole point.
B • Machines	B1 B2	Week 1 • lesson 1.2	--- / 2	Day 1.2. Move Guess My Rule up to Tuesday.
C • Two-step rules	C1 C2	Week 1 • lessons 1.3 and 1.4	--- / 2	Nothing. If C2 is right he has order of operations from Mission 03 — say so out loud, it links the year together.
D • Letters	D1 D2	Week 2 • lessons 2.1 and 2.2	--- / 2	Compress Week 2 to three days and spend the extra two on the trail.
E • Equations	E1 E2	Week 3 • lessons 3.1 and 3.2	--- / 2	Nothing — but move to two-step equations ($2x + 5 = 19$) on Wednesday of Week 3.
F • Meaning	F1 F2	Week 2 • lesson 2.1 • Week 3 • lesson 3.1	--- / 2	Nothing, ever. These two ideas are what the whole mission is for.

BEFORE THE LAST WEEK

Week 4 is the Math Trail and it needs one thing from you: walk it as a genuine student. Get things wrong. Ask him to explain his own ambiguous wording. A capstone he has to defend is worth more than one you praise.

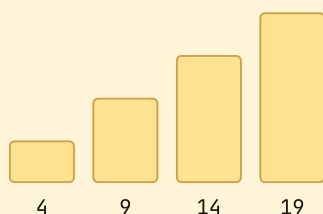
Find the Rule

Extend the pattern, then name the rule in one sentence.

Warm-Up WRITE THE NEXT NUMBER

3, 6, 9, 12
_____5, 10, 15, 20
_____2, 4, 8, 16
_____20, 17, 14, 11
_____1, 4, 7, 10
_____100, 90, 80

WHY A NAMED RULE BEATS COUNTING



Start at 4 and add 5. Every bar is one jump taller than the last, and the jumps are all the same size.

To reach the **tenth** number you add 5 nine times — not ten, because the 4 is already there. $4 + 45 = 49$.

Core

1. Rule $\times 3$, starting at 2. Write the first four numbers.

2. 1, 3, 7, 15, ____ — and say the rule.

3. 4, 9, 14, 19 — the tenth number.

4. 2, 5, 10, 17, ____ — the gaps are not equal. What are they doing?

5. A pattern starts at 6 and adds 5. Write its twentieth number.

★ Challenge NOT GRADED

C1. Write a rule that turns 3 into 10 and 5 into 16. Then test it on 7.
.....

C2. Two patterns start at 1. One adds 4, the other multiplies by 2. Which is bigger at the fifth number, and does the answer change by the tenth?
.....



AXIOM: Say the rule out loud before you write anything. If you can't say it, you haven't found it — you've just guessed the next number correctly.

Function Machines

In goes a number, out comes a number, and the machine never changes its mind.

Warm-Up ONE-STEP MACHINES

+ 8, in 5

× 6, in 4

− 7, in 20

× 10, in 9

÷ 2, in 18

+ 15, in 15

FILL THE TABLE • THE MACHINE IS $\times 3 + 1$

IN	1	2	3	4	10
OUT	4	7			

Notice the outputs climb in threes while the + 1 stays put.
That jump *is* the multiplier, and spotting it is how you crack a rule on Friday.

Core

1. Machine $\times 3 + 4$. In 6.

.....

2. Machine $\times 5 - 2$. In 7.

.....

3. Machine $\times 2 + 9$. In 12.

.....

4. Machine $\times 3 + 4$ gives out 25. In?

.....

5. A machine turns 2 into 7, 3 into 10 and 4 into 13. Write the rule, then use it on 10.

.....

Challenge

C1. A machine turns 1 into 5, 2 into 9 and 3 into 13. What does it do to 12?

.....

C2. Build a machine that turns any number into itself. Write its rule two different ways.

.....

Two-Step Rules

$\times 3$ then $+ 2$ is not the same machine as $+ 2$ then $\times 3$.

✦ Warm-Up MULTIPLY FIRST, EVERY TIME

$2 \times 4 + 1$

$3 \times 5 + 2$

$4 \times 6 - 4$

$5 \times 3 + 5$

$(2 + 4) \times 3$

$2 + 4 \times 3$

SAME TWO STEPS, TWO DIFFERENT MACHINES • PUT IN 4

$4 \rightarrow \times 3 \rightarrow 12 \rightarrow + 2 \rightarrow 14$

Multiply first. The $+ 2$ is added once, at the end.

$4 \rightarrow + 2 \rightarrow 6 \rightarrow \times 3 \rightarrow 18$

Add first and the 2 gets multiplied too — which is where the extra 4 comes from.

◆ Core

$1. \times 4 + 3, \text{ in } 7$

.....

$2. + 3 \text{ then } \times 4, \text{ in } 7$

.....

$3. \times 2 + 6, \text{ in } 9$

.....

$4. \times 6 - 5, \text{ in } 8$

.....

5. Is there any input where $\times 3 + 2$ and $+ 2$ then $\times 3$ agree? Try three inputs before you answer.

.....

★ Challenge

C1. A machine is $\times 2 + 3$. Feed its own output back in. Starting at 1, write the first four numbers you get.

.....

C2. Write a two-step rule that turns 5 into 5 but does not leave every number alone.

.....

Working Backwards

Given the output, undo the machine — last step first.

Warm-Up WHAT UNDOES WHAT

undo + 7

undo $\times 5$ undo $- 4$ undo $\div 3$

+ 6 gave 13, in?

 $\times 4$ gave 28, in?

THE MACHINE $\times 3 + 1$ GAVE OUT 22. RUN IT IN REVERSE.

out 22 $\rightarrow$ take off the 1 $\rightarrow$ 21
21 $\rightarrow$ divide by 3 $\rightarrow$ in 7

Forwards you multiply then add. Backwards you subtract then divide — the opposite operations, in the opposite order. Get the order wrong and 22 becomes $7\frac{1}{3}$ instead of 7.

Core FIND THE INPUT

1. $\times 4 - 1$ gave 272. $\times 5 + 3$ gave 383. $\times 2 + 9$ gave 334. $\div 2 + 4$ gave 12

5. A machine $\times 3 + 5$ gave out 5. What went in, and why is that answer surprising?

Challenge

C1. Write “undo the machine $\times 3 + 1$ ” as an equation with a letter in it, then solve it for an output of 22.

C2. Which machines cannot be undone? Try $\times 0$ and see what goes wrong.

Guess My Rule

Five inputs to crack the machine. Choose them well and you'll need two.

How to play: one person writes a secret rule of the form $\times \text{ ____ } + \text{ ____ }$ and keeps it covered. The other calls out inputs, one at a time, and records the outputs. Crack the rule in five inputs or fewer and you win the round. Then swap.

THE STRATEGY, IN TWO MOVES

Ask for 1, then 2

The jump between those two outputs is the multiplier, on its own, with the add cancelled out.

Then work out the add

Multiply 1 by what you found. The difference between that and the real output is the add.

◆ Play four rounds

ROUND	INPUTS I CHOSE	OUTPUTS	THE RULE	INPUTS USED
1				
2				
3				
4				

★ Challenge

C1. Which single input tells you the least about a rule, and why?

C2. Invent a rule your opponent cannot crack in five inputs. Then explain why it isn't a fair machine.

Weeks 2–4

Outlined, not written out. Week 1's five worksheets are the model.

Week 2 • Variables & Expressions OUTLINED

The letter is not a mystery and not a label — it is a number that hasn't been named yet.

MON 2.1

What a letter is

n is a seat a number sits in. $3n$ is three of them.

TUE 2.2

Evaluate it

Substitute, then work out. One step at a time.

WED 2.3

Write the expression

"Five more than double n " becomes $2n + 5$.

THU 2.4

Two letters

$4a + b$. Substitute both before simplifying.

FRI

Perimeter as a rule

Any square's perimeter is $4s$. Test it on three.

Week 3 • One-Step Equations QUIZ FRIDAY

Equations are scales that must stay level. Whatever you do to one side you do to the other.

MON 3.1

Keep it level

$x + 7 = 19$. Take 7 from both sides.

TUE 3.2

Multiply and divide

$3x = 21$. Divide both sides by 3.

WED 3.3

Check your answer

Substitute it back. Both sides match or you're not done.

THU 3.4

Word to equation

Turn a real situation into an equation, then solve it.

FRI

Mid-unit quiz

Page 10. Eight items, 85% to advance.

Week 4 • The Math Trail CAPSTONE + UNIT TEST

Ten stops through the house, one problem per stop in his handwriting, every mission of the year represented somewhere.

MON 4.1

Design the trail

Map it first. Ten stops, numbered, on paper.

TUE 4.2

Write the problems

Each stop gets a problem and a hidden answer key.

WED 4.3

Mom walks it

He marks her work. Ambiguous wording gets rewritten.

THU

Year-end sweep

All eight error journals. Name the habit that fixed itself.

FRI

Mission 08 test

Pages 11–12, plus the completed trail.

Mid-Unit Quiz

Eight items from Weeks 1 to 3. 85% to keep flying — that's 7 out of 8.

1. What comes next: 6, 11, 16, 21, ____ ?

.....

2. A machine is $\times 4 + 2$. Put in 9.

.....

3. The same machine gives out 30. What went in?

.....

4. If $n = 7$, what is $3n - 4$?

.....

5. Find x : $x + 18 = 45$

.....

6. Find x : $7x = 63$

.....

7. A pattern starts at 3 and adds 4. What is its tenth number?

.....

8. A machine turns 2 into 9, 3 into 13 and 4 into 17. Write its rule, then check your rule on 4 in writing.

.....

.....

Marking note: on items 5 and 6, a right answer with no working still scores — but ask him to show the both-sides step out loud before you move on. That step is what makes two-step equations possible next week.

Unit Test • Part 1

Items 1–8. No time limit. Check every equation by substituting your answer back in.

1. What comes next: 2, 6, 18, 54, ____ ?

.....

2. A pattern starts at 5 and adds 6. Its twelfth number?

.....

3. Machine $\times 5 + 3$. Put in 8.

.....

4. Machine $\times 6 - 4$ gives out 32. What went in?

.....

5. If $n = 9$, what is $4n + 7$?

.....

6. If $a = 4$ and $b = 6$, what is $3a + 2b$?

.....

7. Find x : $x - 26 = 19$

.....

8. Find x : $9x = 108$

.....

Unit Test • Part 2

Items 9–12, the Big Question, and the trail sign-off.

9. Find x : $2x + 5 = 19$

.....

10. A machine turns 5 into 23 and 8 into 35. What does it do to 10?

.....

11. Write “seven more than triple a number” as an expression, then find its value when the number is 8.

.....

12. A bag of marbles is shared equally between 4 people and each gets 13. Write this as an equation using a letter, then solve it.

.....

EXPLAIN YOUR THINKING • THE BIG QUESTION

What does a letter in a math problem actually stand for?

Answer without using the word “mystery.” Use one equation of your own as the example.

.....

.....

.....

.....

.....

MATH TRAIL SIGN-OFF

Ten stops built • Mom walked it • every problem solvable

STOPS COMPLETED _____ / 10

WALKED BY _____

SIGNED _____

11–12 Gold. The Unknown Badge, and Year One is finished.

9–10 Silver. One reteach day, then the badge.

≤ 8 Bronze. Retest the weak strand only — the trail still counts.

Key A

Pre-assessment, Lesson 1.1 and Lesson 1.2.

Mark the reasoning, not the handwriting. Any correct method counts.

Pre-assessment

A1 • 24 | A2 • start at 4, add 5 | B1 • 21 | B2 • 8

C1 • 17 | C2 • no – 22 the other way | D1 • 15 | D2 • 14

E1 • 8 | E2 • 7 | F1 • both sides are the same amount | F2 • no – 10 and 25

Lesson 1.1 • Find the Rule

Warm-Up • 15 | 25 | 32 | 8 | 13 | 70

Core • 1 • 2, 6, 18, 54 | 2 • 31, double and add one | 3 • 49 | 4 • 26, the gaps go 3, 5, 7, 9 | 5 • 101

C1 • $\times 3 + 1$, which sends 7 to 22. C2 • adding 4 is ahead at the fifth number (17 against 16) and far behind by the tenth (37 against 512) — doubling always wins eventually.

Lesson 1.2 • Function Machines

Warm-Up • 13 | 24 | 13 | 90 | 9 | 30

Table • out 10, 13, 31 | Core • 1 • 22 | 2 • 33 | 3 • 33 | 4 • 7 | 5 • $\times 3 + 1$, so 10 gives 31

C1 • $\times 4 + 1$, so 12 gives 49. C2 • $\times 1$, or $+ 0$ — accept either, and both is better.

Key B

Lesson 1.3, Lesson 1.4 and Friday's game.

Where a question asks him to name the rule, the words matter as much as the number.

Lesson 1.3 • Two-Step Rules

Warm-Up • 9 | 17 | 20 | 20 | 18 | 14

Core • 1 • 31 | 2 • 40 | 3 • 24 | 4 • 43 | 5 • they never agree

Item 5 • the second machine is always 4 more, whatever goes in, because the 2 gets multiplied as well. Accept any answer that shows three inputs and spots the constant gap. C1 • 1, 5, 13, 29. C2 • $\times 2 - 5$ sends 5 to 5 but moves everything else.

Lesson 1.4 • Working Backwards

Warm-Up • - 7 | $\div 5$ | + 4 | $\times 3$ | 7 | 7

Core • 1 • 7 | 2 • 7 | 3 • 12 | 4 • 16 | 5 • 0

Item 5 • zero surprises most students because “nothing went in” still counts as an input. C1 • $3x + 1 = 22$, so $x = 7$. C2 • $\times 0$ destroys the input — every number becomes 0, so there is nothing left to undo.

Friday • Guess My Rule

His own rounds, so no key — but the table should show inputs of 1 and 2 early on, and “inputs used” should fall as the rounds go on. That falling number *is* the learning.

C1 • zero tells you the least about the multiplier, since $\times$ anything leaves it at 0 — though it hands you the add for free. C2 • anything not of the form $\times __ + __$, such as “square it,” or a rule that changes between inputs, which is not a machine at all.

Key C

Mid-unit quiz, unit test, and the year-end journal sheet.

Quiz is out of 8, test out of 12. 85% is 7/8 and 11/12.

Mid-unit quiz

$$1 \cdot 26$$

$$2 \cdot 38$$

$$3 \cdot 7$$

$$4 \cdot 17$$

$$5 \cdot 27$$

$$6 \cdot 9$$

$$7 \cdot 39$$

$$8 \cdot \times 4 + 1 \quad (4 \times 4 + 1 = 17 \checkmark)$$

Unit test

$$1 \cdot 162$$

$$2 \cdot 71$$

$$3 \cdot 43$$

$$4 \cdot 6$$

$$5 \cdot 43$$

$$6 \cdot 24$$

$$7 \cdot 45$$

$$8 \cdot 12$$

$$9 \cdot 7$$

$$10 \cdot 43$$

Items 11, 12 and the Big Question

11 • $3n + 7$, which is 31 when $n = 8$. **12** • $m \div 4 = 13$, so $m = 52$ — accept $4 \times 13 = m$ as an equivalent equation.

Big Question • full marks for any answer saying the letter is a number that has not been named yet, and that the equation is a statement about which number it must be. “It stands for something we can work out” earns the mark; “it stands for anything” does not.

Year One • closing sweep

STUDENT PAGE • LAST ONE

Read all eight error journals. Write the one habit that fixed itself, and the one still worth watching.

MISSION

WHAT I'D TELL MYSELF AT THE START OF IT
